

ANALYSTLAB AFRICA

Experience Lab Internship Programme

HealthConnect Clinic Experience Lab

Week 4 – Initial Analysis Document

Data Analytics Track

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

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1. Introduction and Business Context

HealthConnect Clinic is a fictional healthcare provider used for the AnalystLab Africa Experience Lab. The clinic manages appointment-based services and is experiencing missed appointments, unclear drivers of no-show behaviour, inefficient use of appointment slots, and repetitive patient enquiries. The central project question guiding all tracks is:

“How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?”

This document represents the Week 4 contribution of the Data Analytics track. The role of this track within the broader project is to interrogate the historical appointment data to understand attendance patterns, quantify the scale and drivers of no-shows, and propose measurable KPIs and business questions that can guide deeper analysis in subsequent weeks. This foundational understanding will also support the Data Science track (which will build a predictive no-show model) and the Generative AI track (which will need accurate, data-grounded facts about attendance patterns if such information is ever surfaced to the assistant).

2. Resource Review

The following track-relevant project resources were reviewed as part of Part 1 of the Week 4 assignment:

Resource	Description	Status / Use
HealthConnect_Appointment_Data (5,000 records)	Appointment records covering demographics, booking, appointment history, reminders, distance, waiting time and outcomes.	Loaded and profiled; full data quality assessment completed in the companion Excel workbook.
HealthConnect_Data_Dictionary.xlsx	Field-level definitions for the appointment dataset.	Not yet received; column meanings below are inferred from field names and values observed in the data.
HealthConnect_Clinic_Knowledge_Base.docx	Approved clinic facts, procedures, FAQs, and AI safety rules.	Primarily a Generative AI track resource; reviewed for cross-track context only.

A companion workbook, *HealthConnect_Data_Quality_Assessment.xlsx*, contains the full row-level data behind this analysis on a *Raw_Data* tab, with every check in Section 3.2 below implemented as a live Excel formula on a *DQ_Assessment* tab, plus a *Findings_Summary* tab. That workbook is the primary supporting evidence for this section; the summary below reports its results.

3. Dataset Overview and Data Quality Assessment

3.1 Dataset Overview

The dataset contains 5,000 appointment records across 18 substantive fields, one row per scheduled appointment:

Category	Fields (as named in the source file)
Identifiers	appointment_id, patient_id
Patient demographics	gender, age, age_group
Appointment details	appointment_type, appointment_date, appointment_day, appointment_time
Booking information	booking_date, booking_lead_days
Previous appointment history	previous_appointments, previous_no_shows
Reminder information	reminder_sent, reminder_channel
Distance to clinic	distance_to_clinic_km
Waiting time	waiting_time_minutes
Appointment outcome	appointment_outcome (Attended / No-Show / Cancelled) — target field for attendance analysis

The source file also contained two additional, unlabeled columns beyond these 18 fields (see Section 3.2, Finding 6). These are excluded from the categories above and have not been used in any of the analysis in this document.

3.2 Data Quality Assessment — Key Findings

The full checklist below was run against all 5,000 records using live formulas in the companion Excel workbook (*Raw_Data* and *DQ_Assessment* tabs), so every figure can be traced back to a formula and re-verified if the data changes.

1. Completeness

Three fields contain missing values: *reminder_channel* (1,366 blanks, 27.3%), *distance_to_clinic_km* (90 blanks, 1.8%), and *waiting_time_minutes* (60 blanks, 1.2%). All other fields, including *appointment_outcome*, are fully populated.

2. Reminder-channel blanks are logical, not an error

Every blank in `reminder_channel` corresponds exactly to a row where `reminder_sent` = 'No' (0 mismatches found), so this missingness is structural rather than a data quality issue. The 90 missing distance values and 60 missing waiting times appear to be genuinely unrecorded and should be flagged rather than silently imputed if used directly in KPI calculations.

3. Duplicates

No duplicate `appointment_id` values were found (0 of 5,000), and no case of the same patient booked into the same appointment date and time slot more than once was found.

4. Logical consistency

Three cross-field logic checks were run and all passed with zero mismatches: `booking_lead_days` correctly equals the gap between `booking_date` and `appointment_date` in every row; `appointment_day` correctly matches the actual weekday of `appointment_date` in every row; and `reminder_channel` is populated if and only if `reminder_sent` = 'Yes'.

5. Categorical consistency and ranges

All eight categorical fields use a small, consistent set of labels with no case or spelling inconsistencies (e.g., gender: Female 2,488 / Male 2,404 / Prefer not to say 108). All six numeric fields fall within plausible ranges with no negative or impossible values — for example age spans 18–80 (average 48.8) and `distance_to_clinic_km` spans 0.5–45 km (average 10.1 km). An IQR-based outlier screen was run on all six numeric fields; none showed implausible extremes, so any flagged 'outliers' reflect the natural spread of the data rather than entry errors.

6. Two unlabeled source columns

The raw file contains two columns beyond the 18 named fields, with no header. One is completely blank across all 5,000 rows; the other is blank except for a single stray numeric value (2423) in the first data row, with no accompanying label or context. Both have been preserved unedited in the companion workbook and excluded from all analysis pending confirmation from the data owner or Data Dictionary.

7. Class balance

`appointment_outcome` is reasonably balanced: No-Show 48.5% (2,423), Attended 46.3% (2,314), Cancelled 5.3% (263). This favours the planned no-show analysis, since it avoids the severe class imbalance that can distort simple rate comparisons.

Overall, the dataset is clean and internally consistent, and is fit for the KPI calculation and exploratory analysis planned for Week 5, once the two unlabeled columns are clarified and a handling approach is agreed for the small number of missing distance/waiting-time values.

4. Relevant Business Questions

The following questions are proposed to guide the analysis of appointment attendance and no-show behaviour:

1. What proportion of scheduled appointments result in a no-show, and how has this varied over the period covered by the data?
2. Which patient demographic factors (age group, gender, area) are most strongly associated with no-shows?
3. Does the lead time between booking and the appointment date influence the likelihood of a no-show?
4. Is there a measurable difference in no-show rates between patients who received a reminder and those who did not?
5. Does distance from the patient's location to the clinic correlate with attendance?
6. Do patients with a history of prior no-shows show a higher likelihood of missing future appointments?
7. Does waiting time on the day of the appointment relate to future attendance behaviour?

5. Potential KPIs

At this stage, the following KPIs are proposed and justified by relevance to the business questions above. In line with the Week 4 scope, they are identified and linked here only — they will be calculated, analysed and visualised in the next phase.

#	Proposed KPI	Linked business question	Relevance
1	Overall No-Show Rate (%)	Q1	Establishes the baseline scale of the attendance problem the clinic is trying to solve.
2	No-Show Rate by Demographic Segment (age band / gender / area)	Q2	Identifies which patient groups are most at risk of missing appointments, informing targeted interventions.

#	Proposed KPI	Linked business question	Relevance
3	No-Show Rate by Booking Lead-Time Bucket	Q3	Tests whether appointments booked far in advance are more likely to be missed than short-notice bookings.
4	Reminder Effectiveness (No-show rate: reminder sent vs not sent)	Q4	Quantifies whether the clinic's existing reminder process is measurably reducing no-shows.
5	No-Show Rate by Distance Band	Q5	Assesses whether travel distance is a meaningful barrier to attendance, relevant to access-related interventions.

6. Initial Analysis Approach

The proposed approach for the next phase of the Data Analytics track follows a standard analytical workflow:

Data loading and validation — load the CSV in Python (pandas), cross-check columns and types against the Data Dictionary.

Data cleaning — handle missing values, standardise categorical labels, correct data types, and resolve any logical inconsistencies found in Section 3.2.

Exploratory Data Analysis (EDA) — univariate summaries of each variable, followed by bivariate analysis against the appointment outcome (e.g., no-show rate by age band, gender, reminder status).

KPI calculation — compute the five proposed KPIs and document baseline values.

Segmentation and correlation analysis — explore combinations of factors (e.g., distance and lead time together) associated with elevated no-show risk.

Visualisation — build supporting charts (bar charts for categorical comparisons, histograms/boxplots for distributions, trend lines if date range permits) using matplotlib/seaborn, with a dashboard-ready summary planned for a later stage if useful to the project.

Documentation — record findings, assumptions and any data caveats in the analysis notebook and GitHub repository README.

Primary tools: Python (pandas, NumPy, matplotlib) for exploration and cleaning, with SQL considered for any future querying against a structured database version of the data, and Power

BI/Tableau reserved for a dashboard deliverable in a later week, consistent with the “not expected to build a complete dashboard” guidance for Week 4.

7. Assumptions, Limitations, Risks and Dependencies

7.1 Assumptions

- The dataset and its variables are fictional and anonymized but are assumed to reflect realistic, internally consistent appointment patterns for analytical purposes.
- The Data Dictionary is assumed to be the authoritative source for field names, codes and permitted values, and will take precedence over the variable categories inferred in Section 3.1.
- A single appointment outcome field (e.g., Attended / No-show / Cancelled) is assumed to exist and to be usable as the basis for no-show analysis.

7.2 Limitations

- This document was prepared without direct access to the dataset and data dictionary files, so the dataset overview and data quality assessment in Section 3 are a structured plan rather than measured results; they must be re-verified against the real file.
- The size, time span, and completeness of the dataset are not yet known, which limits how confidently trends (e.g., seasonal patterns) can be identified.
- As a fictional/synthetic dataset, patterns may not fully mirror real-world clinic behaviour, which should be kept in mind if findings are ever generalised beyond this project.

7.3 Risks

- Class imbalance between ‘Attended’ and ‘No-show’ outcomes could make simple rate comparisons misleading if not paired with appropriate statistical context.
- Missing or inconsistent values in key fields (reminder status, distance, prior no-show history) could weaken the reliability of the proposed KPIs if not addressed during cleaning.
- Even anonymized patient data should be handled with care; any shared outputs (charts, summaries) should avoid re-identifying small patient subgroups.

7.4 Dependencies

- Timely access to the actual HealthConnect_Appointment_Data.csv and HealthConnect_Data_Dictionary.xlsx files to validate and complete Section 3.
- Alignment with the Data Science track on shared definitions (e.g., how cancellations are treated) so that analytics findings and the future predictive model remain consistent.
- Mentor confirmation of Google Drive/GitHub access for submission.

8. Week 4 Project Summary

1. Problem addressed by this track

Understanding the scale and drivers of patient appointment no-shows at HealthConnect Clinic through exploratory data analysis of historical appointment records, and defining measurable KPIs to track attendance performance.

2. Resources used

HealthConnect business scenario and assignment brief; the described structure of the HealthConnect_Appointment_Data.csv and HealthConnect_Data_Dictionary.xlsx (pending direct file access for full validation).

3. Key observations from Week 4

The dataset (5,000 records, 18 fields) is well-suited to no-show analysis and, on inspection, is clean and internally consistent: no duplicate appointment IDs, zero mismatches on all three cross-field logic checks, and a reasonably balanced outcome split (No-Show 48.5%, Attended 46.3%, Cancelled 5.3%). Three fields have missing values (reminder_channel, distance_to_clinic_km, waiting_time_minutes), and the reminder_channel gaps are fully explained by reminder_sent = 'No'. Two unlabeled columns in the source file (one blank, one with a single stray value) still need clarification before use.

4. Proposed approach

A standard analytics workflow — load, validate, clean, explore, calculate KPIs, segment, visualise, and document — using Python (pandas, matplotlib) as the primary toolset, described in Section 6. The data quality assessment itself was completed in Excel with live formulas so it remains auditable and re-runnable.

5. Key considerations affecting the project

The two unlabeled columns and the small pockets of missing distance/waiting-time data should be resolved with the data owner or Data Dictionary before Week 5; otherwise no major data quality risks were found.

6. Proposed focus for Week 5

Calculate the five proposed KPIs against the validated dataset, begin bivariate analysis to answer the business questions defined in Section 4, and confirm the treatment of the two unlabeled columns and the missing distance/waiting-time values.